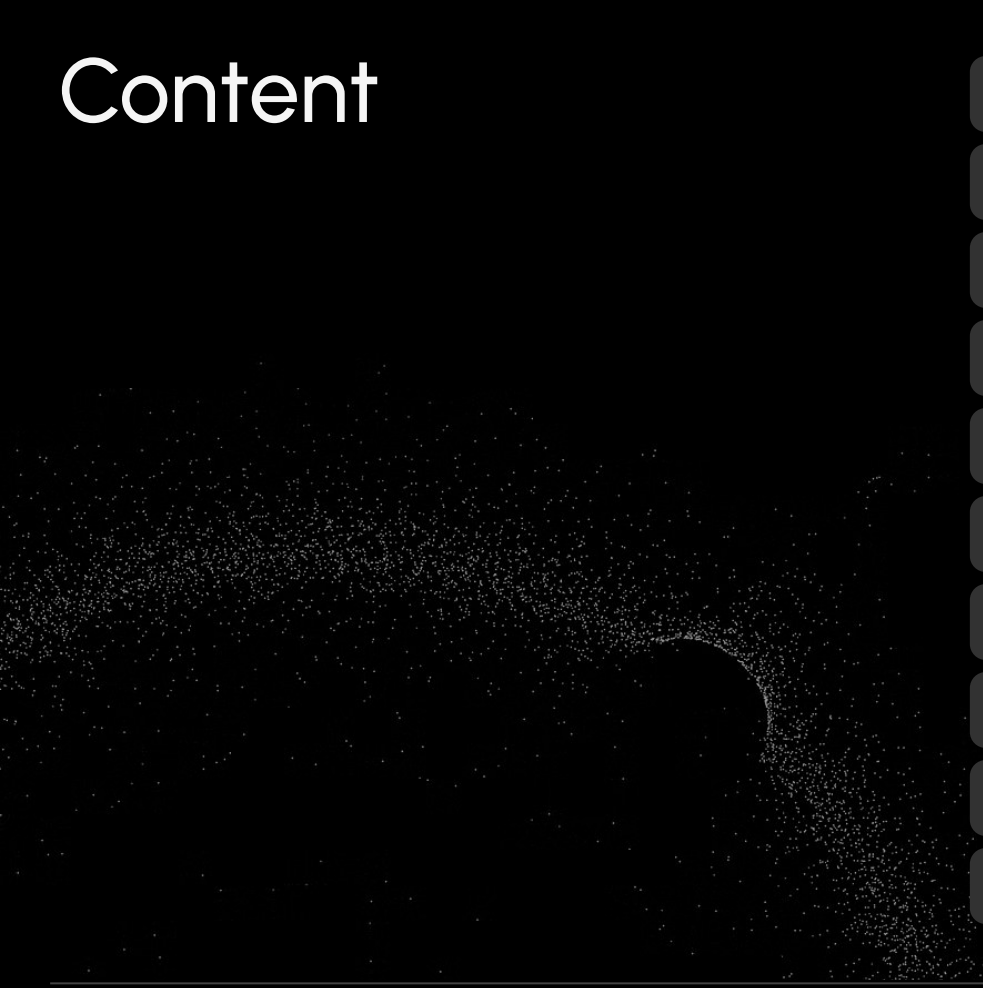
Milestone 1: Presentation

18-755: Networks In The Real World

NoC Topology Characterization Under Diverse Traffic Workloads

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Content



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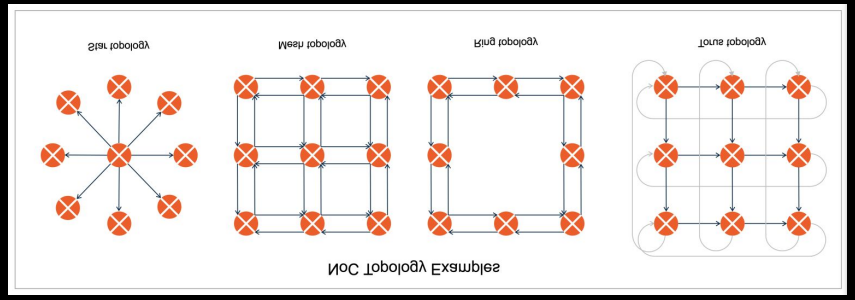
6 Timeline

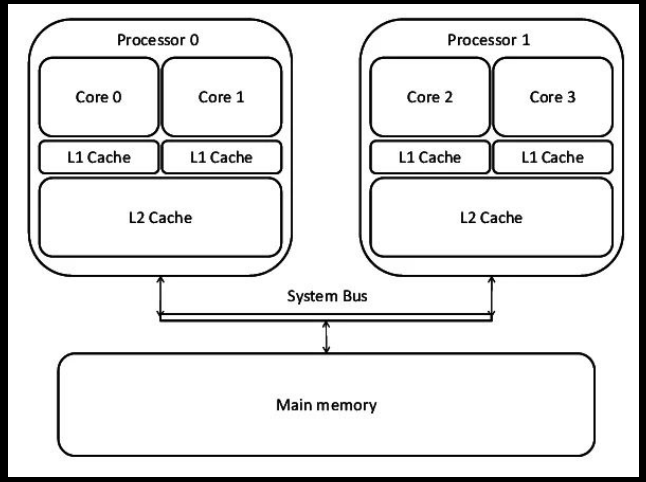
7 Division of Labor

8 Conclusion

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Which structural properties of a NoC topology determine the latency and throughput under different traffic workloads?

- Network on Chip (NoC): **Communication networks** used to connect subsystems in **modern multicore processors**
- On-chip communication becomes a major factor in **performance** due to increasing **number of compute units**
- Different traffic pattern favor different topology

Prior Work

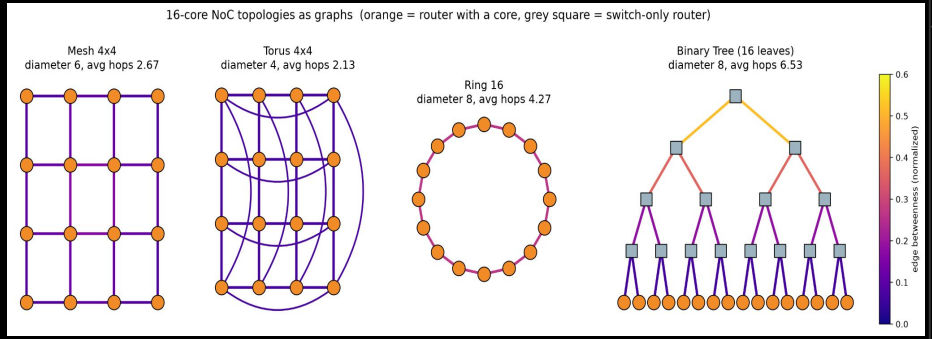
Bakhoda et al. "Throughput-Effective On-Chip Networks for Manycore Accelerators"

- Traffic pattern creates bottleneck in a GPU NoC
- GPU workloads (Many-to-few-to-many):
  - Many cores send request to few memory controllers (MC)
  - Few MCs send reply back to every requesting core
- Cores → MC: Requests are small bytes
- MC → Cores: Replies are much bigger bytes
- Focuses on 2D mesh and one traffic pattern only

Pande et al. "Performance evaluation and design trade-offs for network-on-chip interconnect architectures"

- Throughput and latency vary among different topologies under different workloads
- Uniform traffic: Highly connected topologies
  - More throughput
  - Maintain low latency as sending rate increased
- Localized traffic: Less connected topologies caught up in both condition
- Explored 2 types of traffic: localized and uniform
- No graph metrics used

Our Network: Routers, Links, and Traffic



	Mesh	Torus	Ring	Tree
Routers	16	16	16	31
Links	24	32	16	30
Diameter (hops)	6	4	8	8
Avg hops between cores	2.67	2.13	4.27	8.53

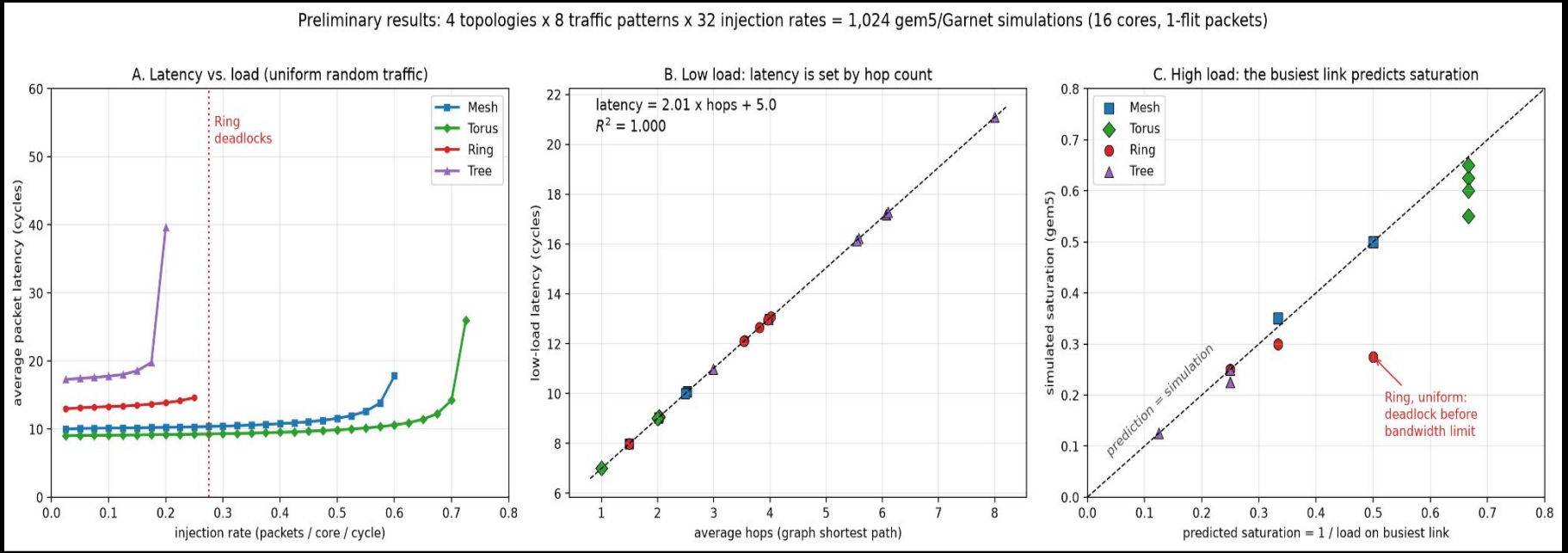
- **Nodes** = routers. Each router connects one core and its slice of memory.
  - Tree also has 15 extra "switch-only" routers that just forward traffic.
- **Links** = two-way on-chip wires: 1 packet per cycle, 1-cycle delay.
- **4 topologies, 16 cores each:** Mesh 4×4 · Torus 4×4 · Ring · Binary Tree
- Input data = synthetic traffic from gem5/Garnet:
  - *Traffic pattern* = who sends to whom (8 patterns: uniform random, transpose, bit-complement, neighbor, …)
  - *Injection rate* = how often each core sends (0.025 → 0.8 packets/cycle)

Predict from the Graph, Then Check in the Simulator

Approach: Main Steps

1. **Build** each topology in gem5/Garnet (we wrote Ring, Torus, Tree)
2. **Analyze the graph** (Python + NetworkX)
   a. Diameter, average hops, betweenness, connectivity
   b. **Prediction:** latency ∝ hops; network saturates when its **busiest link** fills up
3. **Simulate** every topology × traffic pattern × injection rate
4. 4 × 8 × 32 = **1,024 gem5 runs**
5. Record latency, hops, packets delivered, link usage
6. **Compare** prediction vs. simulation → explain why each topology wins or loses
7. **Connect to real chips** (Intel mesh, AMD chiplets, Google TPU torus)

Preliminary Results



- **Low load:** latency = 2 × hops + 5 cycles for every topology and pattern (R^2 = 1.00)
- **High load:** busiest-link prediction matches gem5 within ~5% (20 cases)
- **Ring deadlocks** at 0.275 packets/cycle, half its bandwidth limit (0.5)

Timeline

1

10/20-10/26

gem5 and Garnet setup and implement the scripts needs including for sweep, graph-metric, parser and topology scripts

2

10/27-11/2

Sweep scripts and collect all the necessary data

3

11/3-11/17

Analysis for the data and modify script/rerun if necessary

4

11/17-11/30

Real-world case study relating back to our results and finalize slides/reports

Division of Labor

Steve Cho

- Implementing
  - Graph-metric script
  - stats.txt parser script
- Hypothesize what the results will show
- Analyze the Mesh and Torus topologies from the data from the sweep and graph-metric script

Kushal Agrawal

- Implementing
  - Sweep script
- Hypothesize what the results will show
- Analyze the Ring and Tree topologies from the data from the sweep and graph-metric script

Both

- Real-world case study based on the corresponding analyzed topologies
- Presentation and Report

Conclusion

Main Question

How do the structural properties of different NoC topologies affect latency and throughput under different traffic workloads?

Approach

Simulate Mesh, Torus, Ring and Tree in gem5 and Garnet under 8 traffic patterns from Garnet and measure the graph properties for each

Preliminary Result

- Built Ring, Torus and Tree in gem5 and ran 1,024 simulations
- Low load: latency = 2 × hops + 5 cycles ($R^2 = 1.00$)
- High load: the busiest link predicts saturation within ~5% (the Ring deadlocks early)

Remaining Tasks

- Implement the remaining topologies and run the sweep script
- Measure graph properties and analyze why each topology produced its results
- Real-world case study

References

- A. Bakhoda, J. Kim, and T. M. Aamodt, "Throughput-effective on-chip networks for manycore accelerators," in *Proc. 43rd Annu. IEEE/ACM Int. Symp. Microarchitecture (MICRO)*, 2010, pp. 421–432.
- P. P. Pande, C. Grecu, M. Jones, A. Ivanov, and R. Saleh, "Performance evaluation and design trade-offs for network-on-chip interconnect architectures," *IEEE Trans. Comput.*, vol. 54, no. 8, pp. 1025–1040, 2005.
- **gem5 –** N. Binkert et al., "The gem5 simulator," *ACM SIGARCH Comput. Archit. News*, vol. 39, no. 2, pp. 1–7, 2011.
- **GARNET –** N. Agarwal, T. Krishna, L.-S. Peh, and N. K. Jha, "GARNET: A detailed on-chip network model inside a full-system simulator," in *Proc. ISPASS*, 2009, pp. 33–42.
- W. J. Dally and B. Towles, *Principles and Practices of Interconnection Networks*. Morgan Kaufmann, 2004.
- W. J. Dally and C. L. Seitz, "Deadlock-free message routing in multiprocessor interconnection networks," *IEEE Trans. Comput.*, vol. C-36, no. 5, pp. 547–553, May 1987.
- A. A. Hagberg, D. A. Schult, and P. J. Swart, "Exploring network structure, dynamics, and function using NetworkX," in *Proc. 7th Python in Science Conf. (SciPy)*, 2008, pp. 11–15.
- N. Enright Jerger, T. Krishna, and L.-S. Peh, *On-Chip Networks*, 2nd ed. Morgan & Claypool, 2017.

Image Citation

- Slide 3 Image 1: Stevens , Ashley. "NOCs and the Transition to Multi-Die Systems Using Chiplets." EDN, 2 Aug. 2024, www.edn.com/nocs-and-the-transition-to-multi-die-systems-using-chiplets/.
- Slide 3 Image 2: Rehman, Junaid. "Advantages and Disadvantages of Multi-Core Processors." IT Release, 27 Mar. 2024, itrelease.com/2020/07/advantages-and-disadvantages-of-multi-core-processors/.

Thank you

